Experiment 3

SQL Built-in Functions and Views
Employee Database | Oracle Database 21c XE

1. Source Code

```
-- ------------------------------------------------------------
-- 3a. Conversion, string and date functions
-- ------------------------------------------------------------

-- 3a-0. Create the EMPLOYEES table and insert the data
CREATE TABLE employees (
  employee_id NUMBER(5) PRIMARY KEY,
  first_name  VARCHAR2(20),
  last_name   VARCHAR2(20),
  gender      CHAR(1),
  job_id      VARCHAR2(15),
  department  VARCHAR2(30),
  salary      NUMBER(8,2),
  commission  NUMBER(5,2),
  hire_date   DATE,
  city        VARCHAR2(20)
);
INSERT INTO employees VALUES (101, 'John', 'Smith', 'M', 'IT_PROG', 'IT', 65000, 5, TO_DATE('15-JAN-2020','DD-MON-YYYY
INSERT INTO employees VALUES (102, 'Anita', 'Sharma', 'F', 'HR_REP', 'HR', 52000, 3, TO_DATE('10-JUN-2019','DD-MON-YYY
INSERT INTO employees VALUES (103, 'Rahul', 'Kumar', 'M', 'SA_REP', 'Sales', 48000, 8, TO_DATE('25-AUG-2021','DD-MON-Y
INSERT INTO employees VALUES (104, 'Priya', 'Reddy', 'F', 'MK_MAN', 'Marketing', 72000, 10, TO_DATE('05-MAR-2018','DD-
INSERT INTO employees VALUES (105, 'David', 'Wilson', 'M', 'FI_ACCOUNT', 'Finance', 58000, NULL, TO_DATE('18-DEC-2017'
INSERT INTO employees VALUES (106, 'Sneha', 'Patel', 'F', 'IT_PROG', 'IT', 69000, 6, TO_DATE('12-NOV-2022','DD-MON-YYY
INSERT INTO employees VALUES (107, 'Amit', 'Verma', 'M', 'SA_REP', 'Sales', 45000, 4, TO_DATE('20-JUL-2023','DD-MON-YY
INSERT INTO employees VALUES (108, 'Kiran', 'Rao', 'M', 'HR_REP', 'HR', 50000, NULL, TO_DATE('09-FEB-2021','DD-MON-YYY
INSERT INTO employees VALUES (109, 'Lakshmi', 'Nair', 'F', 'IT_PROG', 'IT', 76000, 7, TO_DATE('14-SEP-2016','DD-MON-YY
INSERT INTO employees VALUES (110, 'Arjun', 'Singh', 'M', 'MK_MAN', 'Marketing', 68000, 5, TO_DATE('30-APR-2019','DD-M
COMMIT;

-- ------------------------------------------------------------
-- Conversion functions
-- ------------------------------------------------------------

-- Q1. Employee ID, first name and hire date as DD-MON-YYYY (TO_CHAR)
SELECT employee_id, first_name,
       TO_CHAR(hire_date,'DD-MON-YYYY') AS hire_date
FROM employees;

-- Q2. Employee ID, first name and salary with a currency symbol (TO_CHAR)
SELECT employee_id, first_name,
       TO_CHAR(salary,'$99,999.99') AS salary
FROM employees;

-- Q3. Add 5000 to each salary (TO_NUMBER)
SELECT employee_id, first_name,
       salary + TO_NUMBER('5000') AS new_salary
FROM employees;

-- Q4. Employees hired after 01-JAN-2020 (TO_DATE)
SELECT * FROM employees
WHERE hire_date > TO_DATE('01-JAN-2020','DD-MON-YYYY');

-- ------------------------------------------------------------
-- String functions
-- ------------------------------------------------------------

-- Q5. Full name using the || operator
SELECT first_name || ' ' || last_name AS full_name
FROM employees;

-- Q6. Full name using CONCAT
SELECT CONCAT(CONCAT(first_name,' '), last_name) AS full_name
FROM employees;

-- Q7. First name left-padded with * (LPAD)
SELECT first_name, LPAD(first_name,10,'*') AS lpad_name
FROM employees;
-- Q8. First name right-padded with * (RPAD)
```

```
SELECT first_name, RPAD(first_name,10,'*') AS rpad_name
FROM employees;

-- Q9. Remove leading spaces (LTRIM)
SELECT first_name, LTRIM(first_name) AS ltrim_name
FROM employees;

-- Q10. Remove trailing spaces (RTRIM)
SELECT last_name, RTRIM(last_name) AS rtrim_name
FROM employees;

-- Q11. First names in lowercase (LOWER)
SELECT LOWER(first_name) AS lower_name FROM employees;

-- Q12. First names in uppercase (UPPER)
SELECT UPPER(first_name) AS upper_name FROM employees;

-- Q13. First names in proper case (INITCAP)
SELECT INITCAP(first_name) AS proper_name FROM employees;

-- Q14. Length of each first name (LENGTH)
SELECT first_name, LENGTH(first_name) AS name_length
FROM employees;

-- Q15. First three characters of the first name (SUBSTR)
SELECT first_name, SUBSTR(first_name,1,3) AS first_three
FROM employees;

-- Q16. Position of the character 'a' in the first name (INSTR)
SELECT first_name, INSTR(first_name,'a') AS position_of_a
FROM employees;

-- ------------------------------------------------------------
-- Date functions
-- ------------------------------------------------------------

-- Q17. Current system date with employee details (SYSDATE)
SELECT employee_id, first_name, last_name,
       SYSDATE AS today
FROM employees;

-- Q18. Next Monday after the hire date (NEXT_DAY)
SELECT first_name, hire_date,
       NEXT_DAY(hire_date,'MONDAY') AS next_monday
FROM employees;

-- Q19. Hire date plus six months (ADD_MONTHS)
SELECT first_name, hire_date,
       ADD_MONTHS(hire_date,6) AS after_6_months
FROM employees;

-- Q20. Last day of the hire month (LAST_DAY)
SELECT first_name, hire_date,
       LAST_DAY(hire_date) AS month_end
FROM employees;

-- Q21. Months worked by each employee (MONTHS_BETWEEN)
SELECT first_name, hire_date,
       ROUND(MONTHS_BETWEEN(SYSDATE,hire_date),2) AS months_worked
FROM employees;

-- Q22. Smaller of salary and 60000 (LEAST)
SELECT first_name, salary,
       LEAST(salary,60000) AS least_value
FROM employees;

-- Q23. Greater of salary and 60000 (GREATEST)
SELECT first_name, salary,
       GREATEST(salary,60000) AS greatest_value
FROM employees;

-- Q24. First day of the hire month (TRUNC)
SELECT first_name, hire_date,
       TRUNC(hire_date,'MM') AS month_start
FROM employees;

-- Q25. Hire date rounded to the nearest month (ROUND)
SELECT first_name, hire_date,
```

```
       ROUND(hire_date,'MONTH') AS rounded_month
FROM employees;

-- Q26. Hire date as DAY, DD-MON-YYYY (TO_CHAR)
SELECT first_name,
       TO_CHAR(hire_date,'DAY, DD-MON-YYYY') AS hire_day
FROM employees;

-- Q27. Employees hired before 01-JAN-2019 (TO_DATE)
SELECT * FROM employees
WHERE hire_date < TO_DATE('01-JAN-2019','DD-MON-YYYY');

-- ------------------------------------------------------------
-- 3b. Creation and dropping of views
-- ------------------------------------------------------------

-- V1. Create view EMP_VIEW (all columns)
CREATE VIEW emp_view AS
SELECT * FROM employees;

-- V2. Create view EMP_BASIC
CREATE VIEW emp_basic AS
SELECT employee_id, first_name, last_name, department, salary
FROM employees;

-- V3. Display all records from EMP_VIEW
SELECT * FROM emp_view;

-- V4. Create view IT_EMPLOYEES
CREATE VIEW it_employees AS
SELECT * FROM employees
WHERE department = 'IT';

-- V5. Create view HIGH_SALARY (salary > 60000)
CREATE VIEW high_salary AS
SELECT * FROM employees
WHERE salary > 60000;

-- V6. Create view HYDERABAD_EMP
CREATE VIEW hyderabad_emp AS
SELECT * FROM employees
WHERE city = 'Hyderabad';

-- V7. Create view FEMALE_EMP
CREATE VIEW female_emp AS
SELECT * FROM employees
WHERE gender = 'F';

-- V8. Create view RECENT_EMPLOYEES (hired on or after 01-JAN-2020)
CREATE VIEW recent_employees AS
SELECT * FROM employees
WHERE hire_date >= TO_DATE('01-JAN-2020','DD-MON-YYYY');

-- V9. Employee ID, first name and salary from HIGH_SALARY
SELECT employee_id, first_name, salary FROM high_salary;

-- V10. Replace EMP_BASIC by adding the CITY column
CREATE OR REPLACE VIEW emp_basic AS
SELECT employee_id, first_name, last_name, department, salary, city
FROM employees;

-- V11. Read-only view EMP_SALARY_VIEW
CREATE VIEW emp_salary_view AS
SELECT employee_id, first_name, last_name, salary
FROM employees
WITH READ ONLY;

-- V12. View SALES_EMP with CHECK OPTION
CREATE VIEW sales_emp AS
SELECT * FROM employees
WHERE department = 'Sales'
WITH CHECK OPTION;

-- V13. Update the salary of employee 101 through EMP_BASIC
UPDATE emp_basic
SET salary = 75000
WHERE employee_id = 101;
-- V14. Delete employee 107 through EMP_VIEW
```

```
DELETE FROM emp_view
WHERE employee_id = 107;

-- V15. Insert a new employee through EMP_BASIC
INSERT INTO emp_basic
VALUES (111, 'Ravi', 'Kumar', 'IT', 60000, 'Hyderabad');

-- V16. Structure of EMP_BASIC
DESC emp_basic;

-- V17. All records from IT_EMPLOYEES
SELECT * FROM it_employees;

-- V18. HIGH_SALARY employees with salary above 70000
SELECT * FROM high_salary
WHERE salary > 70000;

-- V19. All female employees from FEMALE_EMP
SELECT * FROM female_emp;

-- V20. Names and salaries from HYDERABAD_EMP
SELECT first_name, salary FROM hyderabad_emp;

-- V21. Drop EMP_VIEW
DROP VIEW emp_view;

-- V22. Drop HIGH_SALARY
DROP VIEW high_salary;

-- V23. Drop EMP_BASIC
DROP VIEW emp_basic;

-- V24. Create view HR_EMPLOYEES
CREATE VIEW hr_employees AS
SELECT * FROM employees
WHERE department = 'HR';

-- V25. Create view MARKETING_EMP
CREATE VIEW marketing_emp AS
SELECT employee_id, first_name, department, salary
FROM employees
WHERE department = 'Marketing';

-- V26. Create view TOP_EARNERS (salary > 70000)
CREATE VIEW top_earners AS
SELECT * FROM employees
WHERE salary > 70000;

-- V27. Create view EMP_CITY
CREATE VIEW emp_city AS
SELECT employee_id, first_name, last_name, city
FROM employees;

-- V28. List the views that remain
SELECT view_name FROM user_views
ORDER BY view_name;
```

2. Execution and Output

Output shown is sample output of the programs above (Oracle Database 21c XE, SQL*Plus).

3a. Conversion, string and date functions

```
[3a-0]
Table created.
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
Commit complete.
```

Conversion functions

```
[Q1]

EMPLOYEE_ID FIRST_NAME HIRE_DATE
----------- ---------- -----------
        101 John       15-JAN-2020
        102 Anita      10-JUN-2019
        103 Rahul      25-AUG-2021
        104 Priya      05-MAR-2018
        105 David      18-DEC-2017
        106 Sneha      12-NOV-2022
        107 Amit       20-JUL-2023
        108 Kiran      09-FEB-2021
        109 Lakshmi    14-SEP-2016
        110 Arjun      30-APR-2019

10 rows selected.

[Q2]

EMPLOYEE_ID FIRST_NAME SALARY
----------- ---------- -----------
        101 John        $65,000.00
        102 Anita       $52,000.00
        103 Rahul       $48,000.00
        104 Priya       $72,000.00
        105 David       $58,000.00
        106 Sneha       $69,000.00
        107 Amit        $45,000.00
        108 Kiran       $50,000.00
        109 Lakshmi     $76,000.00
        110 Arjun       $68,000.00

10 rows selected.

[Q3]

EMPLOYEE_ID FIRST_NAME NEW_SALARY
----------- ---------- ----------
        101 John            70000
        102 Anita           57000
        103 Rahul           53000
        104 Priya           77000
        105 David           63000
        106 Sneha           74000
        107 Amit            50000
        108 Kiran           55000
        109 Lakshmi         81000
        110 Arjun           73000

10 rows selected.

[Q4]

EMPLOYEE_ID FIRST_NAME LAST_NAME GENDER JOB_ID  DEPARTMENT SALARY COMMISSION HIRE_DATE CITY
----------- ---------- --------- ------ ------- ---------- ------ ---------- --------- ---------
        101 John       Smith     M      IT_PROG IT          65000          5 15-JAN-20 Hyderabad
        103 Rahul      Kumar     M      SA_REP  Sales       48000          8 25-AUG-21 Chennai
        106 Sneha      Patel     F      IT_PROG IT          69000          6 12-NOV-22 Pune
        107 Amit       Verma     M      SA_REP  Sales       45000          4 20-JUL-23 Delhi
```

```
        108 Kiran      Rao       M      HR_REP  HR         50000          09-FEB-21 Hyderabad

5 rows selected.
```

String functions

```
[Q5]

FULL_NAME
------------
John Smith
Anita Sharma
Rahul Kumar
Priya Reddy
David Wilson
Sneha Patel
Amit Verma
Kiran Rao
Lakshmi Nair
Arjun Singh

10 rows selected.

[Q6]

FULL_NAME
------------
John Smith
Anita Sharma
Rahul Kumar
Priya Reddy
David Wilson
Sneha Patel
Amit Verma
Kiran Rao
Lakshmi Nair
Arjun Singh

10 rows selected.

[Q7]

FIRST_NAME LPAD_NAME
---------- ----------
John       ******John
Anita      *****Anita
Rahul      *****Rahul
Priya      *****Priya
David      *****David
Sneha      *****Sneha
Amit       ******Amit
Kiran      *****Kiran
Lakshmi    ***Lakshmi
Arjun      *****Arjun

10 rows selected.

[Q8]

FIRST_NAME RPAD_NAME
---------- ----------
John       John******
Anita      Anita*****
Rahul      Rahul*****
Priya      Priya*****
David      David*****
Sneha      Sneha*****
Amit       Amit******
Kiran      Kiran*****
Lakshmi    Lakshmi***
Arjun      Arjun*****

10 rows selected.

[Q9]

FIRST_NAME LTRIM_NAME
---------- ----------
John       John
Anita      Anita
Rahul      Rahul
Priya      Priya
David      David
Sneha      Sneha
```

```
Amit       Amit
Kiran      Kiran
Lakshmi    Lakshmi
Arjun      Arjun

10 rows selected.
```

[Q10]

```
LAST_NAME RTRIM_NAME
--------- ----------
Smith     Smith
Sharma    Sharma
Kumar     Kumar
Reddy     Reddy
Wilson    Wilson
Patel     Patel
Verma     Verma
Rao       Rao
Nair      Nair
Singh     Singh

10 rows selected.
```

[Q11]

```
LOWER_NAME
----------
john
anita
rahul
priya
david
sneha
amit
kiran
lakshmi
arjun

10 rows selected.
```

[Q12]

```
UPPER_NAME
----------
JOHN
ANITA
RAHUL
PRIYA
DAVID
SNEHA
AMIT
KIRAN
LAKSHMI
ARJUN

10 rows selected.
```

[Q13]

```
PROPER_NAME
-----------
John
Anita
Rahul
Priya
David
Sneha
Amit
Kiran
Lakshmi
Arjun

10 rows selected.
```

[Q14]

```
FIRST_NAME NAME_LENGTH
---------- -----------
John                 4
Anita                5
Rahul                5
Priya                5
David                5
Sneha                5
```

```
Amit                4
Kiran               5
Lakshmi             7
Arjun               5

10 rows selected.
```

[Q15]

```
FIRST_NAME FIRST_THREE
---------- -----------
John       Joh
Anita      Ani
Rahul      Rah
Priya      Pri
David      Dav
Sneha      Sne
Amit       Ami
Kiran      Kir
Lakshmi    Lak
Arjun      Arj

10 rows selected.
```

[Q16]

```
FIRST_NAME POSITION_OF_A
---------- -------------
John                   0
Anita                  5
Rahul                  2
Priya                  5
David                  2
Sneha                  5
Amit                   0
Kiran                  4
Lakshmi                2
Arjun                  0

10 rows selected.
```

Date functions

[Q17]

```
EMPLOYEE_ID FIRST_NAME LAST_NAME TODAY
----------- ---------- --------- ---------
        101 John       Smith     11-OCT-26
        102 Anita      Sharma    11-OCT-26
        103 Rahul      Kumar     11-OCT-26
        104 Priya      Reddy     11-OCT-26
        105 David      Wilson    11-OCT-26
        106 Sneha      Patel     11-OCT-26
        107 Amit       Verma     11-OCT-26
        108 Kiran      Rao       11-OCT-26
        109 Lakshmi    Nair      11-OCT-26
        110 Arjun      Singh     11-OCT-26

10 rows selected.
```

[Q18]

```
FIRST_NAME HIRE_DATE NEXT_MONDAY
---------- --------- -----------
John       15-JAN-20 20-JAN-20
Anita      10-JUN-19 17-JUN-19
Rahul      25-AUG-21 30-AUG-21
Priya      05-MAR-18 12-MAR-18
David      18-DEC-17 25-DEC-17
Sneha      12-NOV-22 14-NOV-22
Amit       20-JUL-23 24-JUL-23
Kiran      09-FEB-21 15-FEB-21
Lakshmi    14-SEP-16 19-SEP-16
Arjun      30-APR-19 06-MAY-19

10 rows selected.
```

[Q19]

```
FIRST_NAME HIRE_DATE AFTER_6_MONTHS
---------- --------- --------------
John       15-JAN-20 15-JUL-20
Anita      10-JUN-19 10-DEC-19
Rahul      25-AUG-21 25-FEB-22
```

```
Priya      05-MAR-18 05-SEP-18
David      18-DEC-17 18-JUN-18
Sneha      12-NOV-22 12-MAY-23
Amit       20-JUL-23 20-JAN-24
Kiran      09-FEB-21 09-AUG-21
Lakshmi    14-SEP-16 14-MAR-17
Arjun      30-APR-19 31-OCT-19

10 rows selected.
```

[Q20]

```
FIRST_NAME HIRE_DATE MONTH_END
---------- --------- ---------
John       15-JAN-20 31-JAN-20
Anita      10-JUN-19 30-JUN-19
Rahul      25-AUG-21 31-AUG-21
Priya      05-MAR-18 31-MAR-18
David      18-DEC-17 31-DEC-17
Sneha      12-NOV-22 30-NOV-22
Amit       20-JUL-23 31-JUL-23
Kiran      09-FEB-21 28-FEB-21
Lakshmi    14-SEP-16 30-SEP-16
Arjun      30-APR-19 30-APR-19

10 rows selected.
```

[Q21]

```
FIRST_NAME HIRE_DATE MONTHS_WORKED
---------- --------- -------------
John       15-JAN-20         80.89
Anita      10-JUN-19         88.05
Rahul      25-AUG-21         61.57
Priya      05-MAR-18        103.21
David      18-DEC-17        105.79
Sneha      12-NOV-22         46.99
Amit       20-JUL-23         38.73
Kiran      09-FEB-21         68.08
Lakshmi    14-SEP-16        120.92
Arjun      30-APR-19         89.41

10 rows selected.
```

[Q22]

```
FIRST_NAME SALARY LEAST_VALUE
---------- ------ -----------
John        65000       60000
Anita       52000       52000
Rahul       48000       48000
Priya       72000       60000
David       58000       58000
Sneha       69000       60000
Amit        45000       45000
Kiran       50000       50000
Lakshmi     76000       60000
Arjun       68000       60000

10 rows selected.
```

[Q23]

```
FIRST_NAME SALARY GREATEST_VALUE
---------- ------ --------------
John        65000          65000
Anita       52000          60000
Rahul       48000          60000
Priya       72000          72000
David       58000          60000
Sneha       69000          69000
Amit        45000          60000
Kiran       50000          60000
Lakshmi     76000          76000
Arjun       68000          68000

10 rows selected.
```

[Q24]

```
FIRST_NAME HIRE_DATE MONTH_START
---------- --------- -----------
John       15-JAN-20 01-JAN-20
Anita      10-JUN-19 01-JUN-19
Rahul      25-AUG-21 01-AUG-21
```

```
Priya      05-MAR-18 01-MAR-18
David      18-DEC-17 01-DEC-17
Sneha      12-NOV-22 01-NOV-22
Amit       20-JUL-23 01-JUL-23
Kiran      09-FEB-21 01-FEB-21
Lakshmi    14-SEP-16 01-SEP-16
Arjun      30-APR-19 01-APR-19

10 rows selected.

[Q25]

FIRST_NAME HIRE_DATE ROUNDED_MONTH
---------- --------- -------------
John       15-JAN-20 01-JAN-20
Anita      10-JUN-19 01-JUN-19
Rahul      25-AUG-21 01-SEP-21
Priya      05-MAR-18 01-MAR-18
David      18-DEC-17 01-JAN-18
Sneha      12-NOV-22 01-NOV-22
Amit       20-JUL-23 01-AUG-23
Kiran      09-FEB-21 01-FEB-21
Lakshmi    14-SEP-16 01-SEP-16
Arjun      30-APR-19 01-MAY-19

10 rows selected.

[Q26]

FIRST_NAME HIRE_DAY
---------- ----------------------
John       WEDNESDAY, 15-JAN-2020
Anita      MONDAY   , 10-JUN-2019
Rahul      WEDNESDAY, 25-AUG-2021
Priya      MONDAY   , 05-MAR-2018
David      MONDAY   , 18-DEC-2017
Sneha      SATURDAY , 12-NOV-2022
Amit       THURSDAY , 20-JUL-2023
Kiran      TUESDAY  , 09-FEB-2021
Lakshmi    WEDNESDAY, 14-SEP-2016
Arjun      TUESDAY  , 30-APR-2019

10 rows selected.

[Q27]

EMPLOYEE_ID FIRST_NAME LAST_NAME GENDER JOB_ID     DEPARTMENT SALARY COMMISSION HIRE_DATE CITY
----------- ---------- --------- ------ ---------- ---------- ------ ---------- --------- ---------
        104 Priya      Reddy     F      MK_MAN     Marketing   72000         10 05-MAR-18 Hyderabad
        105 David      Wilson    M      FI_ACCOUNT Finance     58000            18-DEC-17 Mumbai
        109 Lakshmi    Nair      F      IT_PROG    IT          76000          7 14-SEP-16 Kochi

3 rows selected.
```

3b. Creation and dropping of views

```
[V1]
View created.

[V2]
View created.

[V3]

EMPLOYEE_ID FIRST_NAME LAST_NAME GENDER JOB_ID     DEPARTMENT SALARY COMMISSION HIRE_DATE CITY
----------- ---------- --------- ------ ---------- ---------- ------ ---------- --------- ---------
        101 John       Smith     M      IT_PROG    IT          65000          5 15-JAN-20 Hyderabad
        102 Anita      Sharma    F      HR_REP     HR          52000          3 10-JUN-19 Bengaluru
        103 Rahul      Kumar     M      SA_REP     Sales       48000          8 25-AUG-21 Chennai
        104 Priya      Reddy     F      MK_MAN     Marketing   72000         10 05-MAR-18 Hyderabad
        105 David      Wilson    M      FI_ACCOUNT Finance     58000            18-DEC-17 Mumbai
        106 Sneha      Patel     F      IT_PROG    IT          69000          6 12-NOV-22 Pune
        107 Amit       Verma     M      SA_REP     Sales       45000          4 20-JUL-23 Delhi
        108 Kiran      Rao       M      HR_REP     HR          50000            09-FEB-21 Hyderabad
        109 Lakshmi    Nair      F      IT_PROG    IT          76000          7 14-SEP-16 Kochi
        110 Arjun      Singh     M      MK_MAN     Marketing   68000          5 30-APR-19 Jaipur

10 rows selected.

[V4]
View created.

[V5]
View created.
```

```
[V6]
View created.

[V7]
View created.

[V8]
View created.

[V9]

EMPLOYEE_ID FIRST_NAME SALARY
----------- ---------- ------
        101 John        65000
        104 Priya       72000
        106 Sneha       69000
        109 Lakshmi     76000
        110 Arjun       68000

5 rows selected.

[V10]
View created.

[V11]
View created.

[V12]
View created.

[V13]
1 row updated.

[V14]
1 row deleted.

[V15]
1 row created.

[V16]

 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 EMPLOYEE_ID                               NOT NULL NUMBER(5)
 FIRST_NAME                                         VARCHAR2(20)
 LAST_NAME                                          VARCHAR2(20)
 DEPARTMENT                                         VARCHAR2(30)
 SALARY                                             NUMBER(8,2)
 CITY                                               VARCHAR2(20)

[V17]

EMPLOYEE_ID FIRST_NAME LAST_NAME GENDER JOB_ID  DEPARTMENT SALARY COMMISSION HIRE_DATE CITY
----------- ---------- --------- ------ ------- ---------- ------ ---------- --------- ---------
        101 John       Smith     M      IT_PROG IT          75000          5 15-JAN-20 Hyderabad
        106 Sneha      Patel     F      IT_PROG IT          69000          6 12-NOV-22 Pune
        109 Lakshmi    Nair      F      IT_PROG IT          76000          7 14-SEP-16 Kochi
        111 Ravi       Kumar                   IT          60000                      Hyderabad

4 rows selected.

[V18]

EMPLOYEE_ID FIRST_NAME LAST_NAME GENDER JOB_ID  DEPARTMENT SALARY COMMISSION HIRE_DATE CITY
----------- ---------- --------- ------ ------- ---------- ------ ---------- --------- ---------
        101 John       Smith     M      IT_PROG IT          75000          5 15-JAN-20 Hyderabad
        104 Priya      Reddy     F      MK_MAN  Marketing   72000         10 05-MAR-18 Hyderabad
        109 Lakshmi    Nair      F      IT_PROG IT          76000          7 14-SEP-16 Kochi

3 rows selected.

[V19]

EMPLOYEE_ID FIRST_NAME LAST_NAME GENDER JOB_ID  DEPARTMENT SALARY COMMISSION HIRE_DATE CITY
----------- ---------- --------- ------ ------- ---------- ------ ---------- --------- ---------
        102 Anita      Sharma    F      HR_REP  HR          52000          3 10-JUN-19 Bengaluru
        104 Priya      Reddy     F      MK_MAN  Marketing   72000         10 05-MAR-18 Hyderabad
        106 Sneha      Patel     F      IT_PROG IT          69000          6 12-NOV-22 Pune
        109 Lakshmi    Nair      F      IT_PROG IT          76000          7 14-SEP-16 Kochi

4 rows selected.

[V20]

FIRST_NAME SALARY
---------- ------
```

```
John        75000
Priya       72000
Kiran       50000
Ravi        60000

4 rows selected.

[V21]
View dropped.

[V22]
View dropped.

[V23]
View dropped.

[V24]
View created.

[V25]
View created.

[V26]
View created.

[V27]
View created.

[V28]

VIEW_NAME
---------------
EMP_CITY
EMP_SALARY_VIEW
FEMALE_EMP
HR_EMPLOYEES
HYDERABAD_EMP
IT_EMPLOYEES
MARKETING_EMP
RECENT_EMPLOYEES
SALES_EMP
TOP_EARNERS

10 rows selected.
```